Study: Mach-Zehnder Interferometer Circuit, SOI (DRAFT)

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I. Introduction

PHOTONIC integrated circuits (PICs) are an essential building block for the future of computing. The Mach-Zehnder interferometer (MZI) serves as one of the fundamental components on PICs, allowing for a single source of light to be self-interfered or for the interference of two distinct beams of light, utilising the relative phase shift between them. In general, the optical path length between the two arms of the interferometer are different and sometimes variable. The most simple way to accomplish this is to vary the physical length between the paths while keeping the refractive index constant. For variability, either electro-optic or thermo-optic modulators can be used.

Using the aforementioned modulators, MZIs can be constructed using 2x2 splitters and configured in such a way to create an optical NxN switch, allowing the efficient transfer of data around a PIC. In this report, for simplicity and to suit the scale of the assignment, we consider a 3x3 switch with static path length differences rather than variable optic modulators.

II. Theory

To construct an MZI, waveguides are used to direct the light towards two paths of different optical lengths. The paths are then recombined and the light interaction which occurs depends on the relative phases. For the basic theory of an MZI, consider another fundamental component, the Y-Branch.

The Y-Branch comes in two designs which are similar but distinct; the splitter and the combiner. The splitter has one path divided into two. If the intensity of the input is I_i, each output takes half of the intensity, $I_1 = I_2 = I_i/2$. For the electric field, $I_i \propto E^2$. Therefore at the output:

$$E_1 = E_i/\sqrt{2} \tag{1}$$

Similar theory applies to the combiner, where the output electric field is given by:

$$E_o = E_1/\sqrt{2} \tag{2}$$

in the case of a single input, or

$$E_o = \frac{E_1 + E_2}{\sqrt{2}} \tag{3}$$

in the case where both inputs are used.

Now consider the case where first light is split using a y-branch splitter, propagates along waveguides with a difference in length ΔL, then combined using a y-branch coupler. Electric field propagation, in general, is described by the plane wave equation:

$$E = E_i e^{i(\omega t - \beta z)} \tag{4}$$

where $\beta = \frac{2\pi n_{eff}}{\lambda}$ is the propagation constant and z is the distance travelled.

In the general case, we consider the two paths at the output of the splitter may have different propagation characteristics (β) and different loss coefficients (α).

For path lengths L_1 and L_2, we write the electric fields at the end of the paths, just before the combiner, as:

$$E_{o1} = E_1 e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} \text{ and} \tag{5}$$

$$E_{o2} = E_2 e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2} \tag{6}$$

The splitter output, therefore, is:

$$E_o = \frac{E_{o1} + E_{o2}}{\sqrt{2}} = \frac{E_i}{2}(e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}) \tag{7}$$

The intensity is then given by:

$$I_o = \frac{I_i}{4}|e^{-i\beta_1 L_1 - \frac{\alpha_1}{2} L_1} + e^{-i\beta_2 L_2 - \frac{\alpha_2}{2} L_2}| \tag{8}$$

If we consider the simpler case of equal propagation constants and no loss:

$$I_o = \frac{I_i}{2}[1 + cos(\beta \Delta L)] \tag{9}$$

For a single output device, the remainder of the light $(1 - cos(\beta \Delta L))$ is radiated away. However, if we consider a 2x2 coupler instead, that light is simply directed towards the other output port.

From this we can calculate the difference in path length required to split the light in the ratio I_o out of the bottom port of the interferometer and $I_i - I_o$ out of the top port. A 50/50 split ($I_0 = \frac{I_i}{2}$) at 1550nm in silicon, for example, will yield a ΔL as follows:

$$\Delta L = \frac{arccos(\frac{2I_0}{I_i} - 1)}{\beta} \tag{10}$$

$$\Delta L = \frac{\pi}{2\beta} \tag{11}$$

$$\Delta L = 155nm \tag{12}$$

III. Waveguide Analysis

The propagation constant mentioned previously can vary depending on the physical structure of the waveguide. Lumerical Mode software can be used to calculate various properties of waveguides including propagation of different components of electric and magnetic fields, propagation of different modes (TE and TM) and values for effective indices.

For SOI (Silicon on Insulator) waveguides, a common standard for 1550nm light is a 500nm wide by 220nm tall waveguide. The electric field intensity for this waveguide is shown in figures 1 and 2 for the first TE and TM modes, respectively.

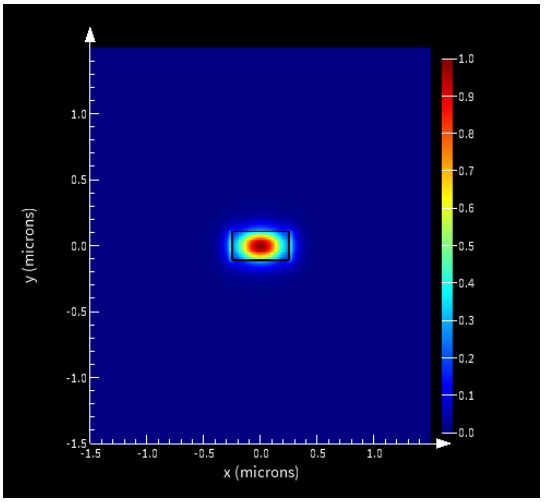


Fig. 1. Electric field intensity of the first TE mode of a 500 x 220 nm silicon waveguide for light at 1550 nm.

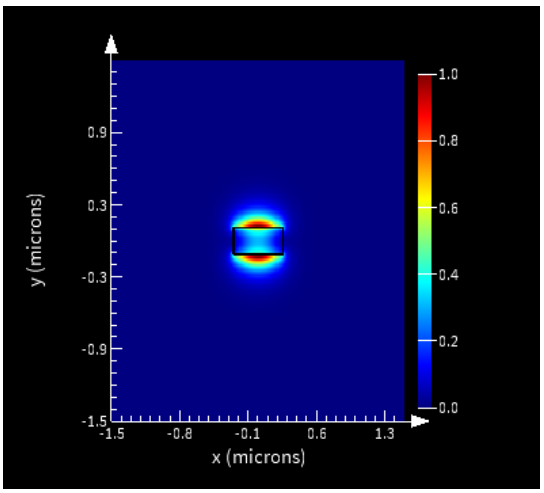


Fig. 2. Electric field intensity of the first TM mode of a 500 x 220 nm silicon waveguide for light at 1550 nm.

If the waveguide is made 100nm narrower or wider, not much of a change in the electric field is seen, however, there is a slight change in effective index (see table ref). More interestingly, if the waveguide is made slightly taller, it can be seen that the first TM mode can now be accommodated inside the waveguide and rather than the bulk of the electric field being outside the waveguide (fig. 2) it is contained and can propagate (fig. 3). This gives insight into the reasoning for the standard size; to isolate only the first TE mode (or only TE modes).

The effective index can be calculated from Lumerical Mode over a range of wavelengths, giving insight into how the waveguide will respond when there are frequency fluctuations in the light source. The index data can also be used to create a "compact model" of the waveguide, which is useful for transferring a strong approximation of the waveguide behaviour more easily. The compact model in general is a Taylor expansion of the effective index around a center wavelength, with the following format:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \tag{13}$$

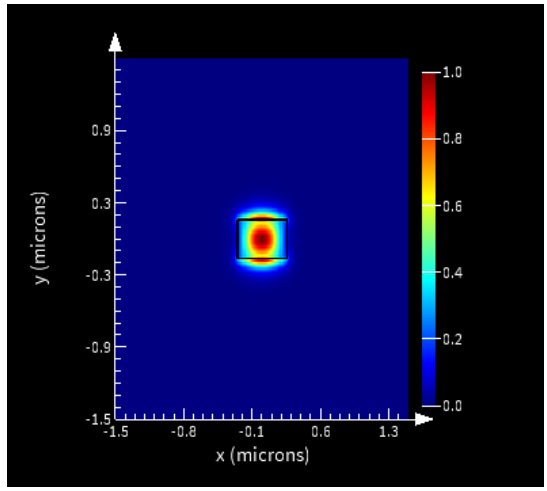


Fig. 3. Electric field intensity of the first TM mode of a 500 x 320 nm silicon waveguide for light at 1550 nm.

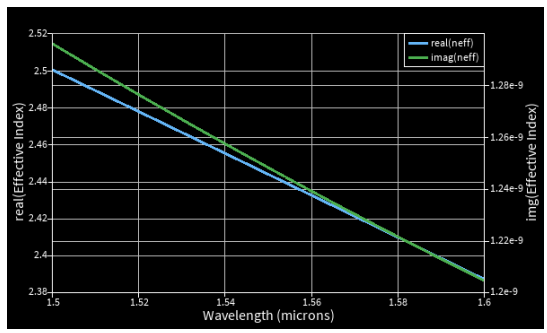


Fig. 4. The real and imaginary components of the effective index for the first TE mode of the 500 x 220 nm waveguide over a range of wavelengths. Note the y-axis scale for the imaginary component is on the right hand side.

Using a matlab script, the data from lumerical can be fitted to the compact model equation, giving the compact model equation for the 500 x 220 nm waveguide:

$$n_{eff}(\lambda) = 2.44391 + 1.13177(\lambda - 1.55) - 0.0426968(\lambda - 1.55)^2 \tag{14}$$

where the wavelengths are given in microns.

Lumerical Mode calculates many other important properties of the waveguide under simulation alongside the effective index, such as the group index (5), the loss per centimetre and the dispersion. While for very short and simple circuits, these values may not have large effects, for more complex projects these waveguide attributes must be carefully considered.

IV. Mach-Zehnder Interferometer

In this section, we consider MZIs with 2 inputs and 2 outputs, therefore composed of two 2x2 couplers with waveguides of differing lengths in between them. The couplers add some phase along the coupling length during crossover. This added phase can affect the output of the MZI, so to mitigate this, the second coupler is flipped horizontally and vertically. This has the effect of adding phase to the bottom path in the first coupler and adding the same amount of phase to the top path in the second coupler, negating the additions. We can then base our MZI output calculations on equation 9.

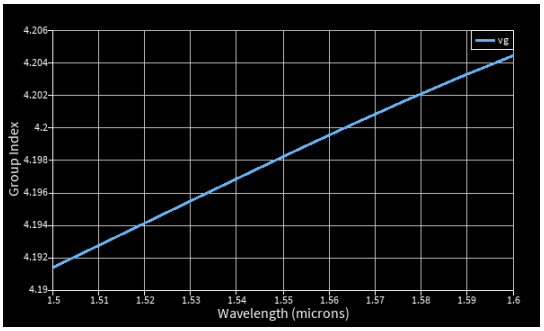


Fig. 5. The group index of the 500 x 220 nm waveguide.

An example of the gain for both of the outputs for an MZI with $\Delta L = 100nm$ is shown below in figure 6. As the gain of one port increases, the other decreases, leading to opposite maximums and minimums. The peaks are regularly spaced; the distance between them is called the free spectral range (FSR). The FSR of an MZI can be determined by the following equation:

$$FSR = \frac{\lambda^2}{n_g \Delta L} \tag{15}$$

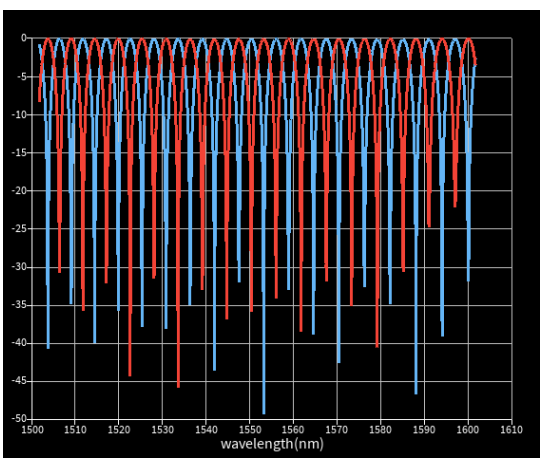


Fig. 6. The gain at both of the output ports of and imbalanced MZI with $\Delta L = 100nm$ over a broad spectrum.

For the purposes of a 3x3 splitter, we would ideally like a flat response to accommodate a broader range of wavelengths. To achieve this, we minimize ΔL, making the FSR as large as possible. Examining the transfer function equation (equation 9), the maximum is reached when the cosine term is equal to 1, and the minimum when that term is -1. This tells us the argument of the cosine term, which is the change in phase, functionally ranges from 0 to π (i.e. this range covers all possible power splits between ports). This in turn gives us the fundamental range of values for $\Delta L : [0, \sim 320]$. To minimize the FSR, ideally only very small differences in wavelength will be used, however, there are generally manufacturing constraints limiting this. In the most ideal case, we would additionally add phase changes to the end of the circuit to line up the phases of all three outputs.

V. Interconnect and KLayout Implementation

The 3×3 (1-to-3) optical switch was modeled in *Lumerical INTERCONNECT* to validate the circuit logic and subsequently translated to a physical mask in *KLayout*.

A. Circuit Simulation and Spectral Analysis

To ensure phase symmetry and robust routing, a symmetric MZI architecture was adopted within the INTERCONNECT environment. As shown in Fig. 7, the circuit utilizes cascaded stages of 2×2 bidirectional couplers (BDCs). To mitigate the intrinsic phase shift added during crossover events—which was measured in the ONA at approximately 1.39 rad for a standalone BDC—the second coupler in each interferometer stage is flipped both horizontally and vertically. This configuration ensures that the intrinsic coupling phase is neutralized by adding phase to the bottom path in the first coupler and an equal amount to the top path in the second, allowing the output intensity to be governed primarily by the intentional physical path length difference ΔL.

The simulation utilized an Optical Network Analyzer (ONA) to verify the phase relationship across the C-band. By maintaining the total phase shift within the $[0, \pi]$ range, the device covers all possible power splits between ports.

The spectral response was analyzed to distinguish between device performance and fiber-to-chip coupling effects. The raw gain response exhibits a parabolic profile characteristic of the grating couplers. By de-embedding the grating coupler response, the intrinsic circuit transmission is revealed to be remarkably flat across the C-band (Fig. 8). This broadband performance is a direct result of minimizing ΔL, which maximizes the Free Spectral Range (FSR) and ensures functionality over a wide operating window.

B. Physical Layout

The physical translation in *KLayout*, illustrated in Fig. 9, adheres to the 500×220 nm SOI waveguide standard. The static phase shifts are implemented through localized waveguide bends that create precise length differences. For this design, the fundamental range of ΔL values $[0, \sim 320]$ nm was utilized to achieve the required phase shifts at a center wavelength of 1550 nm.

To ensure a flat spectral response and accommodate a broader range of wavelengths, ΔL was minimized to maximize the Free Spectral Range (FSR). The layout accounts for specific bend radii to minimize radiation loss while ensuring that the first TM mode remains poorly confined compared to the primary TE mode.

References

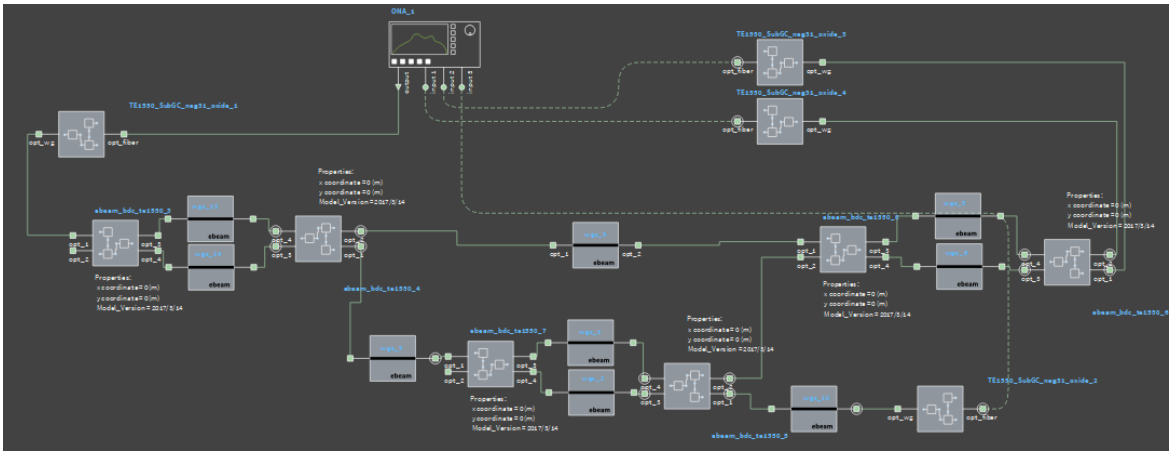
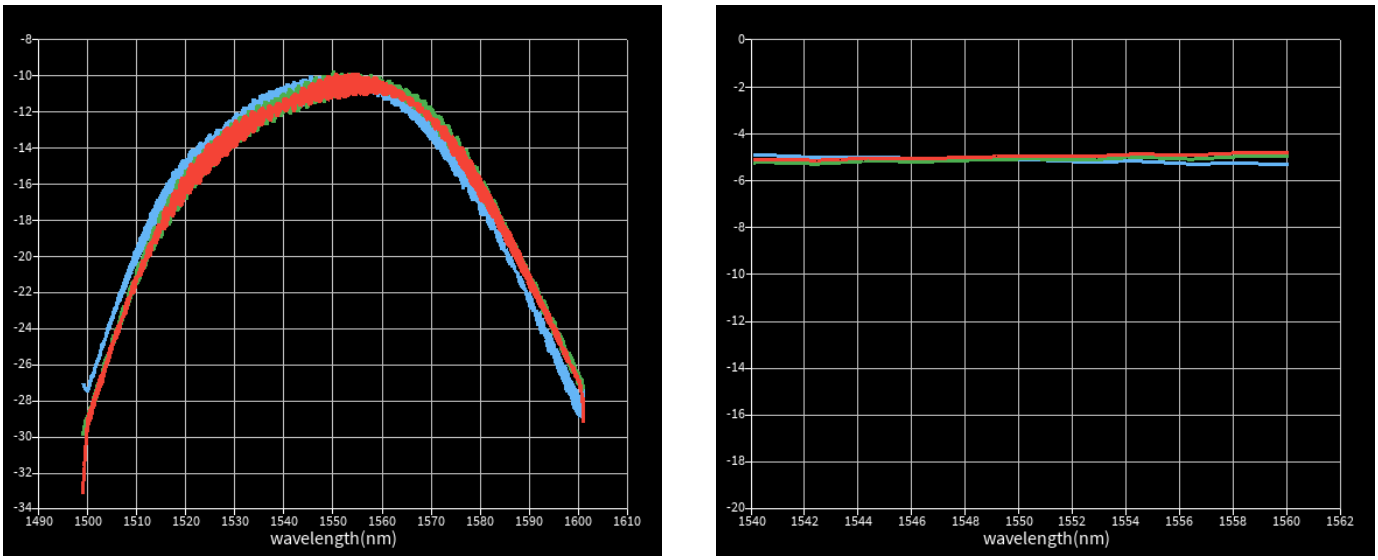


Fig. 7. Full INTERCONNECT circuit schematic of the 1-to-3 switch. The cascaded symmetric MZI stages allow for wavelength-independent routing logic by neutralizing coupler-induced phase offsets.



(a) Gain with Grating Couplers

(b) Intrinsic Circuit Gain (De-embedded)

Fig. 8. Spectral response comparison. The raw data (left) shows the Gaussian-like envelope of the grating couplers, while the de-embedded data (right) demonstrates the broadband, flat response of the MZI circuit achieved by maximizing the FSR.

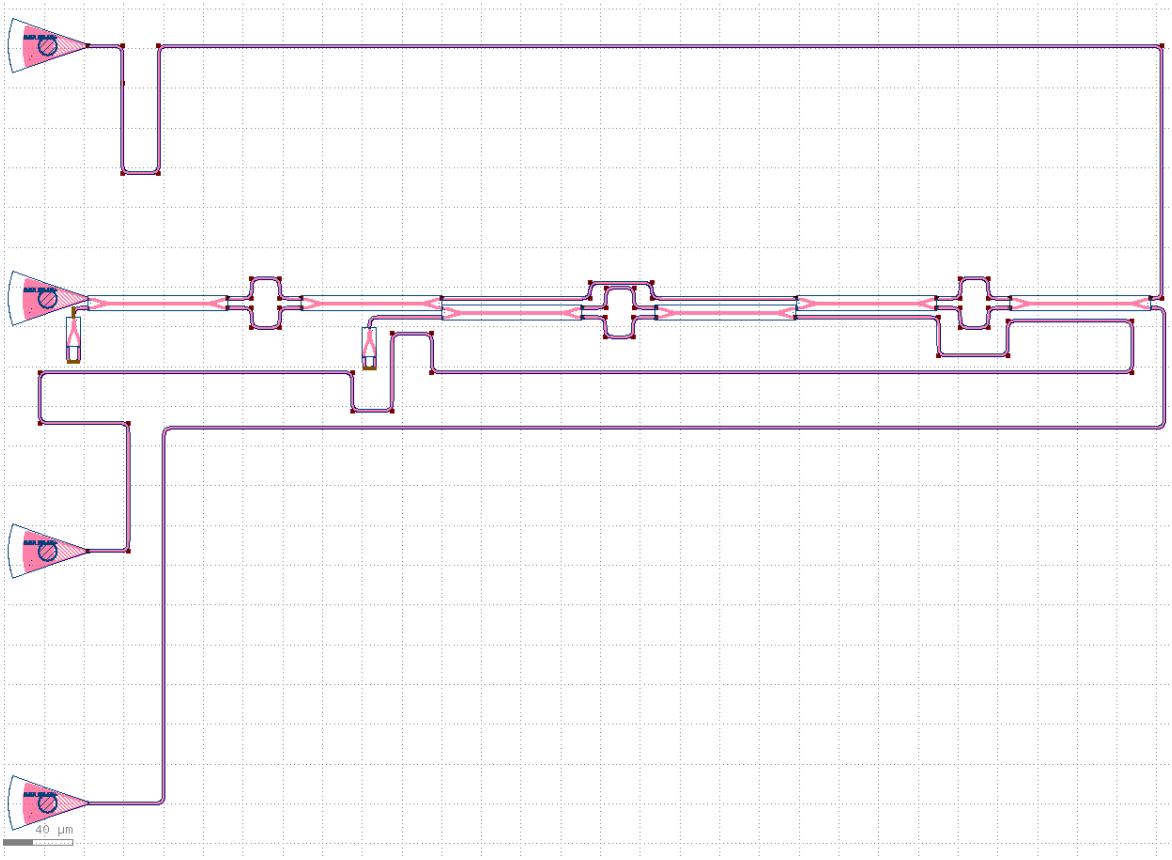


Fig. 9. Complete KLayout GDSII mask design. The layout utilizes 500×220 nm waveguides to isolate the TE mode while precisely controlling arm lengths for static switching.